

ICT702 Data Visualisation

# Data Dashboards

## Module 8



I would like to acknowledge the traditional custodians of the  
land we are meeting on,  
The Gubbi Gubbi/Kabi Kabi peoples,  
and pay my respect to Elders past, present and emerging.



# Part 3

# Espléndido Jugo y Batido, Inc. (EJB) Data Dashboard

A portion of Espléndido Jugo y Batido, Inc. data in the file *EJB*

|   | A        | B        | C        | D          | E            | F              | G                   | H             | I                           | J                           |
|---|----------|----------|----------|------------|--------------|----------------|---------------------|---------------|-----------------------------|-----------------------------|
| 1 | Order ID | Flavor   | Category | \$ Sales   | Date Ordered | Date Delivered | Distribution Center | New Customer? | Service Satisfaction Rating | Product Satisfaction Rating |
| 2 | 92145    | Beet     | Juices   | \$605.97   | 1/1/2018     | 1/3/2018       | ID                  | Yes           | 3                           |                             |
| 3 | 92145    | Apple    | Juices   | \$1,549.00 | 1/1/2018     | 1/3/2018       | ID                  | Yes           | 5                           | 4                           |
| 4 | 92145    | Apple    | Juices   | \$1,986.28 | 1/1/2018     | 1/3/2018       | ID                  | Yes           | 5                           | 4                           |
| 5 | 92145    | Orange   | Juices   | \$16.43    | 1/1/2018     | 1/3/2018       | ID                  | Yes           | 5                           | 3                           |
| 6 | 92145    | Cucumber | Juices   | \$1,594.93 | 1/1/2018     | 1/3/2018       | ID                  | Yes           |                             |                             |
| 7 | 92145    | Pear     | Juices   | \$590.15   | 1/1/2018     | 1/3/2018       | ID                  | Yes           |                             | 5                           |

- EJB manufactures bottled juices and smoothies.
- EJB ships products from seven distribution centers.
- Data orders from the past three years
- Each data record consists of the amount of one of the flavours included in an order
- An order can consist of multiple records in the data

# EJB Dashboard: Project Needs

Data dashboard project needs

- **EJB Goal:** allow the mid-level operations management team to track its sales over the most recent three years.
- **EJB Specific objectives:**
  1. Track dollar sales for each of its distribution centers by year and new and existing customers
  2. Track dollar sales for each category (juices and smoothies) by year, flavour, and month
  3. Monitor time to delivery by distribution center for each year
  4. Generate sales in dollars across category and flavour by distribution center at the year, month and date ordered levels
  5. The data dashboard should automatically update when adding new data.

# EJB Dashboard: Variable Requirements

## Data dashboard KPIs, variables of interest, and required tables and charts

- **Dashboard's KPIs:** total dollar sales and average delivery times.
- **Variables of interest:** month ordered, year ordered, date ordered, category, flavour, distribution center, and whether the customer is new or existing.
- **Chart and table requirements:**
  1. A stacked column chart of total sales across distribution centers and year ordered by a new or existing customer
  2. A line chart of total sales across year and month ordered by a new or existing customer
  3. A clustered column chart of sales across year ordered and flavour by category
  4. A clustered bar chart of average time to delivery across distribution centers by year ordered
  5. A table of the total dollar across category and flavour by distribution center and the date ordered

# EJB Dashboard: Chart 1

PivotChart for total sales with slicers for distribution center, year ordered, and whether the customer is new

Total Sales (Thousand \$) by Distribution Center, Year Ordered, and New Customer

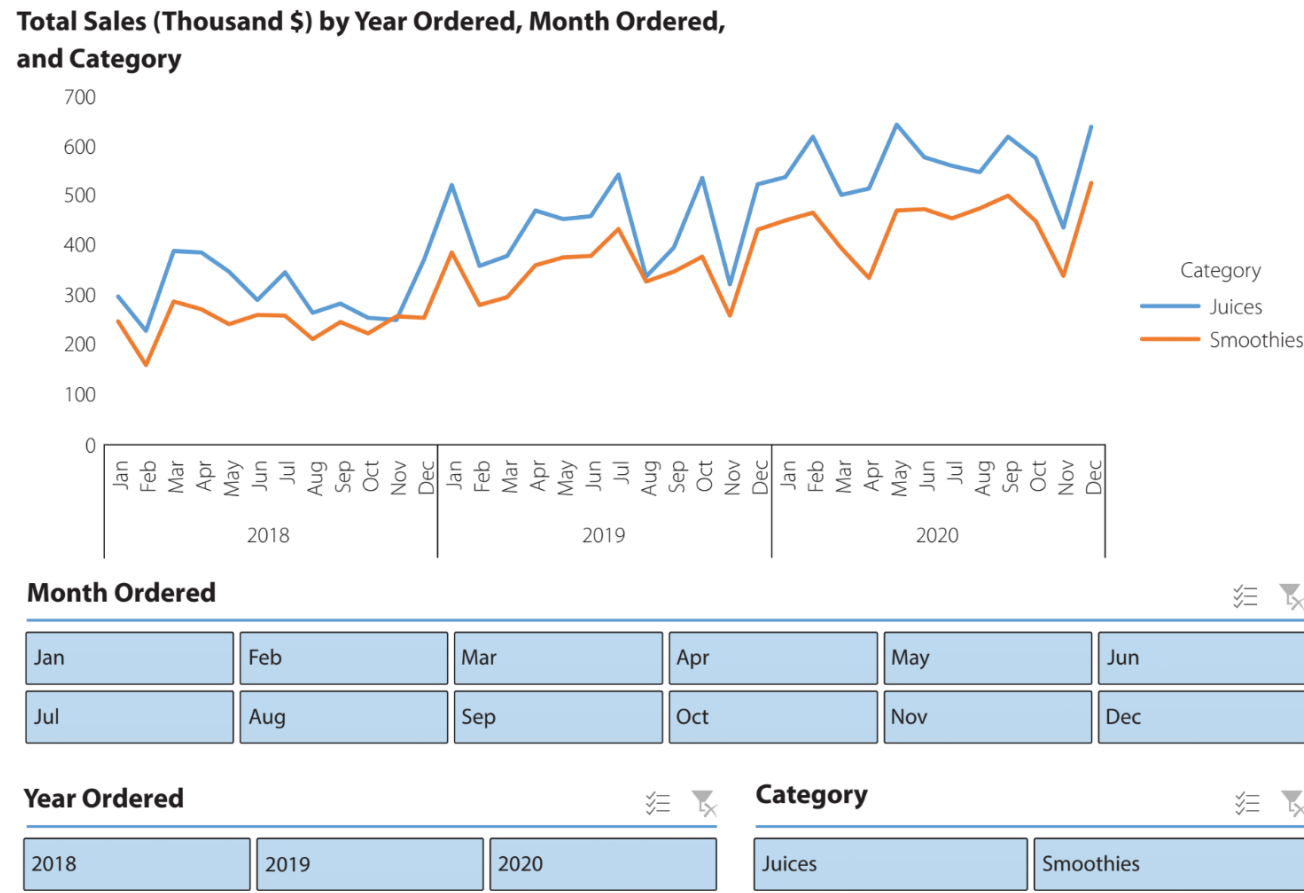


The stacked column chart view on this PivotChart will allow the user to graphically display total dollar sales for any combination of years ordered, distribution centers, and whether or not the customers are new.



# EJB Dashboard: Chart 2

## PivotChart for total sales with slicers for year ordered, month ordered, and category



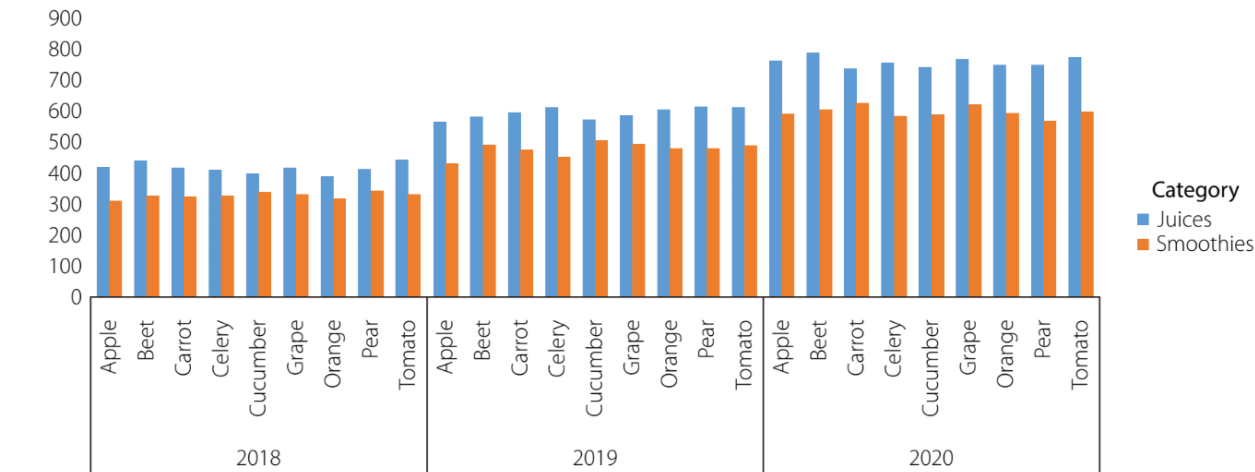
The line chart view on this PivotChart will allow the user to graphically display total dollar sales for any combination of years ordered, months ordered, and category.



# EJB Dashboard: Chart 3

## PivotChart for total sales with slicers for year ordered, flavour, and category

Total Sales (Thousand \$) by Year Ordered, Flavor, and Category



Flavor

|       |        |        |        |          |
|-------|--------|--------|--------|----------|
| Apple | Beet   | Carrot | Celery | Cucumber |
| Grape | Orange | Pear   | Tomato |          |

Year Ordered

|      |      |      |
|------|------|------|
| 2018 | 2019 | 2020 |
|------|------|------|

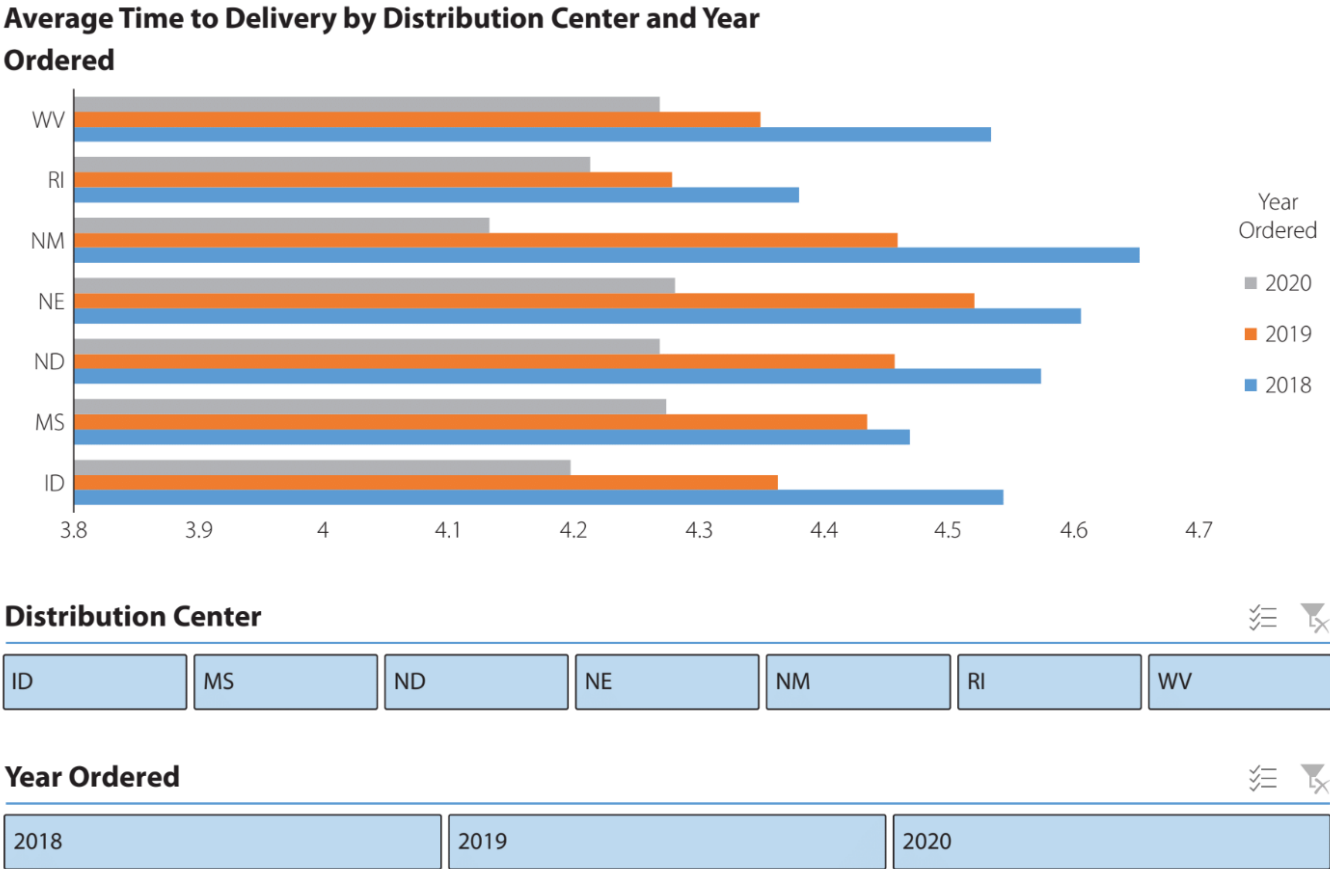
Category

|        |           |
|--------|-----------|
| Juices | Smoothies |
|--------|-----------|

The clustered column chart view on this PivotChart will allow the user to graphically display total dollar sales for any combination of years ordered, flavour, and category.

# EJB Dashboard: Chart 4

Chart 4 - PivotChart for average time to delivery with slicers for distribution center and year ordered



The clustered bar chart view on this PivotChart will allow the user to graphically display the average time to delivery for any combination of distribution centers and years ordered.



# EJB Dashboard: Build a PivotTable

## Steps to build the PivotTable for the EJB data dashboard

**Step 1.** Select any cell in the *EJBData* table of the *Data* worksheet  
Click the **Insert** tab on the Ribbon and select **PivotTable** in the **Tables** group  
Accept the default in the **Create PivotTable** DIALOG BOX  
Click **OK**

**Step 2.** Rename *Dashboard* the newly created worksheet .

**Step 3.** In the **Dashboard** worksheet, click the empty **PivotTable** to open the **PivotTable Fields** pane and drag the fields to the areas as shown to the left.

After autofitting column widths, the final PivotTable is shown next.

Drag fields between areas below:

| FILTERS               | COLUMNS    |
|-----------------------|------------|
| Year Ordered ▼        | Category ▼ |
| Month Ordered ▼       | Σ Values ▼ |
| Day Ordered ▼         |            |
| Distribution Center ▼ |            |

| ROWS     | Σ VALUES               |
|----------|------------------------|
| Flavor ▼ | Sum of \$ Sales ▼      |
|          | Average of Time t... ▼ |
|          |                        |

# EJB Dashboard: PivotTable for the EJB data dashboard

|    | A | B                   | C               | D               | E               | F               | G                     |
|----|---|---------------------|-----------------|-----------------|-----------------|-----------------|-----------------------|
| 1  |   |                     |                 |                 |                 |                 |                       |
| 2  |   |                     |                 |                 |                 |                 |                       |
| 3  |   |                     |                 |                 |                 |                 |                       |
| 4  |   | Year Ordered        | (All)           |                 |                 |                 |                       |
| 5  |   | Month Ordered       | (All)           |                 |                 |                 |                       |
| 6  |   | Day Ordered         | (All)           |                 |                 |                 |                       |
| 7  |   | Distribution Center | (All)           |                 |                 |                 |                       |
| 8  |   |                     |                 |                 |                 |                 |                       |
| 9  |   |                     | Column Labels   |                 |                 |                 |                       |
| 10 |   |                     | Juices          | Average of Time | Smoothies       | Average of Time | Total Sum of \$ Sales |
| 11 |   | Row Labels          | Sum of \$ Sales | to Delivery     | Sum of \$ Sales | to Delivery     | Total Average of Time |
| 12 |   | Apple               | \$1,746,990     | 4.4642          | \$1,330,694     | 4.3665          | \$3,077,684           |
| 13 |   | Beet                | \$1,805,180     | 4.3649          | \$1,418,969     | 4.4392          | \$3,224,149           |
| 14 |   | Carrot              | \$1,748,337     | 4.3631          | \$1,423,311     | 4.4293          | \$3,171,647           |
| 15 |   | Celery              | \$1,774,726     | 4.4234          | \$1,360,251     | 4.2735          | \$3,134,977           |
| 16 |   | Cucumber            | \$1,710,376     | 4.3108          | \$1,429,580     | 4.3693          | \$3,139,956           |
| 17 |   | Grape               | \$1,770,301     | 4.4432          | \$1,441,864     | 4.3115          | \$3,212,164           |
| 18 |   | Orange              | \$1,740,004     | 4.4367          | \$1,387,955     | 4.3809          | \$3,127,959           |
| 19 |   | Pear                | \$1,771,377     | 4.4047          | \$1,389,352     | 4.4240          | \$3,160,728           |
| 20 |   | Tomato              | \$1,823,193     | 4.3985          | \$1,414,450     | 4.3529          | \$3,237,643           |
| 21 |   | Grand Total         | \$15,890,484    | 4.4012          | \$12,596,425    | 4.3719          | \$28,486,909          |
| 22 |   |                     |                 |                 |                 |                 |                       |

# The EJB Data Dashboard: A Potential Layout

When considering the relative positioning, the objective is to minimize eye travel by putting charts likely to be used together adjacent to each other.

Table - Total Sales (Thousand \$) and Average Time to Delivery Across Category and Flavor by Distribution Center and Year, Month, and Day Ordered

Chart 3 - Total Sales (Thousand \$) by Year Ordered, Flavor, and Category

Chart 2 - Total Sales (Thousand \$) by Year Ordered, Month Ordered, and Category

Chart 1 - Total Sales (Thousand \$) by Distribution Center, Year Ordered, and New Customer

Chart 4 - Average Time to Delivery by Distribution Center and Year Ordered

- Chart 1 and Chart 3 display sales history by year
- Chart 1 and Chart 4 display analyses at the distribution center level
- Chart 2 and Chart 3 display sales at category level
- The table of total sales across category and flavour by distribution center and date ordered should be located at the top, for easy access

# The EJB Data Dashboard

The style and settings for any slicers can be changed by:

- selecting the slicer
- clicking **Options** on the ribbon
- using the various tools such as those in the **Slicer Styles** group



# The EJB Data Dashboard with Connected Slicers



The EJB data dashboard conveniently allows the users to filter each chart separately using slicers.

Connected slicers allow using a single slicer to control multiple related charts, such as a single Distribution Center slicer that controls all the relevant charts displayed on the dashboard.

# Final Review of a Data Dashboard

- Test the PivotCharts and PivotTables extensively using combinations of slicers and filters to ensure that:
  - the information is accurate
  - the display is appealing and informative
- Ensure the final data dashboard meets management team's objectives
- Allow the users to extensively test the final data dashboard to ensure the dashboard is:
  - user friendly
  - functions in the manner expected by its users
  - produces the outputs expected and needed by its users
- Work with management team and users to create an ongoing monitoring and revision process

# Common Mistakes in Data Dashboard Design

- Not considering the organization's needs and objectives, such as:
  - neglecting information that is material to the objectives of the organization
  - focusing on information that is not material to the objectives of the organization
  - neglecting the actual users throughout the data dashboard design process
- Not considering the environment(s) for the data dashboard use
- Failing to position charts and tables on the data dashboard properly
- Using inappropriate or ineffective type of graph for data and its message
- Neglecting principles of chart and table design when creating individual components and overall dashboard
- Creating a data dashboard that is either too cluttered or too sparse
- Designing an unattractive visual display
- Not considering the organization and user's potential future needs

# Summary

In Module 8 you have learnt how:

- Data dashboards can help users understand and investigate data and make better decisions.
- Principles and common applications of the various types of data dashboards help build better dashboards.
- To consider the purpose of the data dashboard and the needs of the data dashboard's users.
- To display in the data dashboard the information that is important to the organization.
- To use PivotTables, PivotCharts and slicers to build a data dashboard in Excel.
- To link slicers to multiple PivotTables.
- To protect a data dashboard so users cannot make permanent changes.

# End of Part 3

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